



HMI For Prosthetic Limb

4IER

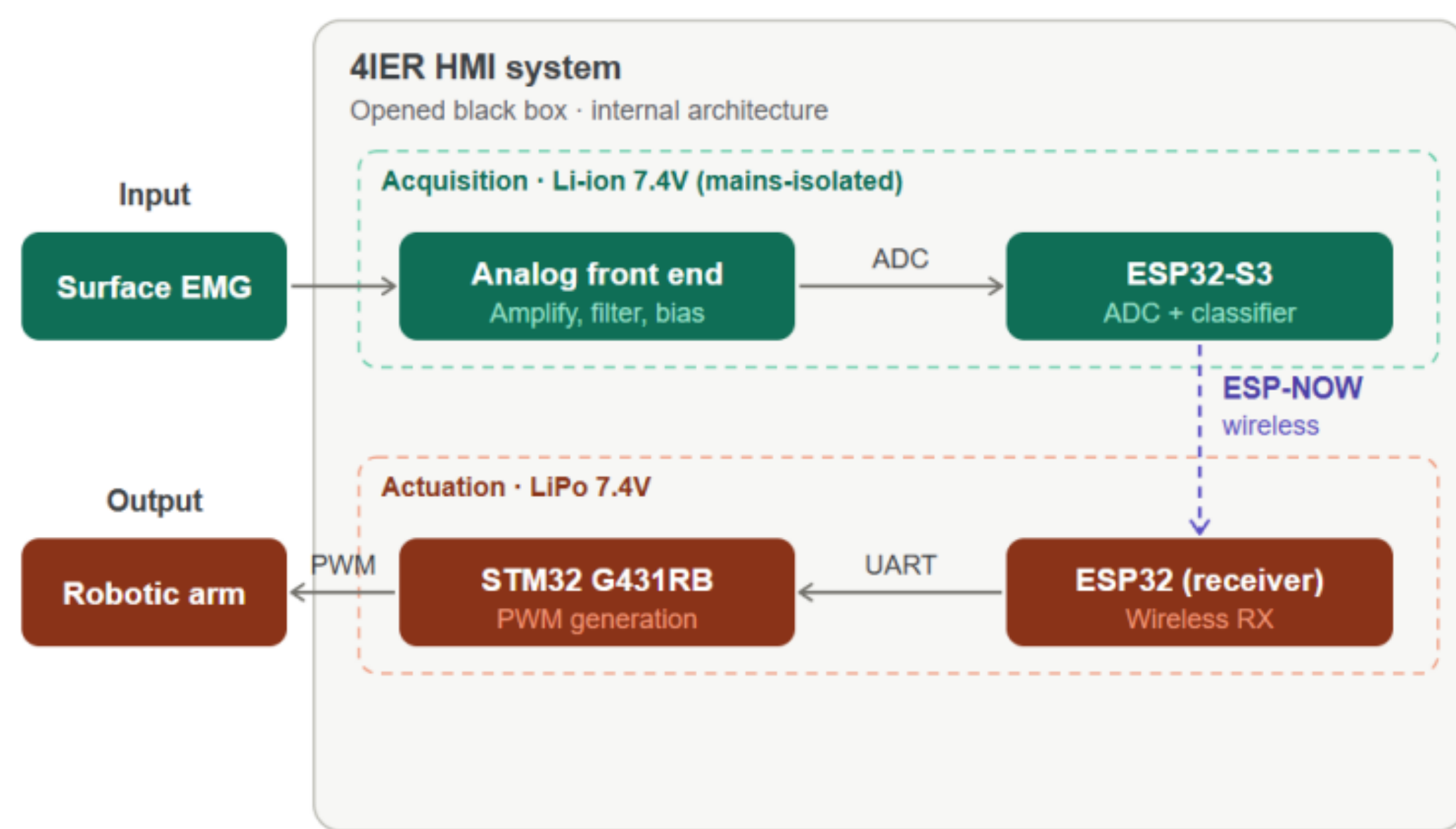
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Project Description

Developed by senior METU Electrical and Electronics Engineering students in team 4IER, this advanced HMI project controls a robotic arm using human muscle signals. The system integrates biosignal data acquisition with intelligent control mechanisms. By leveraging deep neural networks to decode real-time EMG patterns, it allows the robotic arm to operate safely and with extreme precision even in the most sensitive and confined spaces.

System Design



Requirements

- End-to-End Delay:** Total system latency must be bounded between 0 and 200 ms for responsive control.
- Classification Accuracy:** System operational accuracy target spans the 0 – 100% range.
- Leakage Current:** Maximum allowable leakage is capped at 5 μ A to ensure strict patient safety.
- Power Consumption:** The overall operational current draw of the system is constrained to 2A.
- Isolation:** The system must operate as a fully battery-powered floating node.

Compliance with Requirements

Requirement	Required Value	Measured Result	Status
Total End-to-End Delay	0 – 200 ms	180 ms	Success
Classification Accuracy	0 – 100 %	98.65 %	Success
Leakage Current	< 10 μ A	5 μ A	Success
Power Consumption	0–10A	2A	Success
Isolation	Battery-Powered	Battery-Powered	Success

Deliverables

- Universal EMG Acquisition & Classification
- 3D-Printed Robotic Arm System
- Embedded TinyML Gesture Classifier
- End-to-End HMI Control System

Features

- Wireless Communication:** Seamless and ultra-fast data link between the control system and the prosthetic arm using ESP-NOW.
- Autonomous Edge Computing:** Executes all DSP and classification tasks directly on-device, completely eliminating the need for an external PC.
- Decoupled Architecture:** Distinct physical separation between the bio-signal acquisition and actuation stages for maximum patient safety.
- Real-Time Actuation:** Hardware-level PWM generation via STM32 ensures highly precise and smooth robotic movements.
- Scalable Sensor Interface:** Expandable analog front-end architecture, ready to accommodate additional signal channels for more complex gesture recognition.

Test Results & Performance

Parameter	Value	Actual Performance	Expected Performance	Error
Wireless Latency	8.5 ms		< 50 ms	-
Classification Accuracy	96%		> 90%	-
Power Consumption	17W		< 33W	-
Total Delay	180 ms		< 200 ms	-
Leakage Current	5.5 μ A		< 10 μ A	-
Grasping, Carrying, Positioning and Releasing Test	95% for grasping and carrying 80% for positioning and releasing		> 90% accuracy	15% for positioning and releasing

Cost Breakdown

Component	Qty	Price
ESP32-S3 DevKit	1	\$15.00
ESP32 Receiver	1	\$10.00
STM32G431KB	1	\$12.00
sEMG Sensors	3	\$60.00
SG90 Servos	4	\$16.00
Li-Po Battery	2	\$20.00
Other (PCB, wires)	–	\$29.00
Grand Total		\$162.00